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PII: S0045-6535(20)33229-X

DOI: <https://doi.org/10.1016/j.chemosphere.2020.129032>

Reference: CHEM 129032

To appear in: *ECSN*

Received Date: 29 September 2020

Revised Date: 7 November 2020

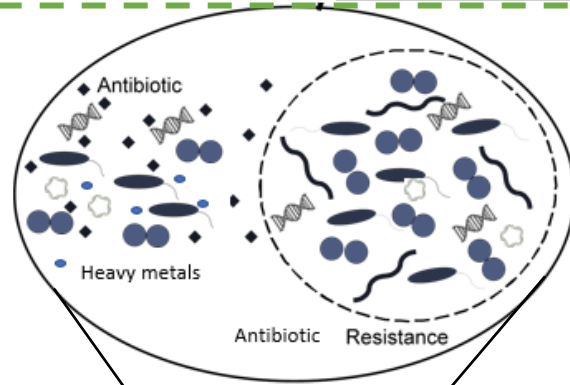
Accepted Date: 17 November 2020

Please cite this article as: Sorinolu, A.J., Tyagi, N., Kumar, A., Munir, M., Antibiotic Resistance Development and Human Health Risks during Wastewater Reuse and Biosolids Application in Agriculture, *Chemosphere*, <https://doi.org/10.1016/j.chemosphere.2020.129032>.

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Wastewater Treatment plants



AR plasmids
AR genes

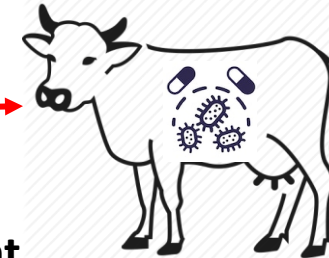
Treated Wastewater

Biosolids

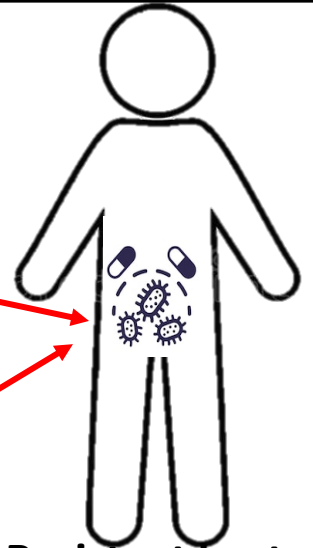
Antibiotic resistance (AR) development
in soil bacteria and plants



Resistant bacteria
in food crops



Resistant bacteria
in food animals



Resistant bacteria
in human gut



Holistic studies are needed for comprehensive Quantitative Microbial Risk Assessment (QMRA)

Antibiotic Resistance Development and Human Health Risks during Wastewater Reuse and Biosolids Application in Agriculture

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Highlights

- Accumulation of antibiotics and antibiotic resistance (AR) determinants in soil and edible plants.
- The roles of contaminants interactions and soil properties on soil resistome and microbiota.
- A framework for a comprehensive AR risks assessment during treated wastewater (TWW) irrigation and biosolid use in agriculture.
- Future directions for quantitative human health AR risk assessment due to TWW and biosolid fertilization.

Abstract

The reuse of treated wastewater (TWW) and sewage sludge are considered as solutions to the limited water resource and sludge disposal issues, respectively. The associated environmental and human health risks need to be analyzed to assess whether they are safe solutions or not. This paper discusses issues that relate to the accumulation of antibiotics and antibiotic resistance (AR) determinants in agricultural lands and crops, following TWW irrigation and biosolid amendment. Exposure assessment and dose-response assessment are the two important aspects of risk assessment discussed in this paper. Finally, research gaps in current knowledge that are relevant to a comprehensive and quantitative AR risk assessment were identified which includes: 1.) Studies on soil conditions that increase the frequency of horizontal gene transfer (HGT) between the native soil resistome and pathogenic microbes in biosolids and TWW 2.) Holistic studies that examine the accumulation or dissipation of antibiotics, antibiotic resistant bacteria (ARB) and antibiotic resistance genes (ARGs) from the irrigation/biosolids application stage to crop consumption stage 3.) The influences of soil environmental conditions (e.g. salinity, nutrients) on the fate of ARB and ARGs in soil and translocation in edible plants 4.) The development of dose-response models for quantifying the risks of infection due to ARB.

Keywords: Wastewater Reuse, Biosolids, Antibiotic Resistance, Horizontal Gene Transfer, Risk Assessment

1. Introduction

Freshwater resources are limited, and meeting the needs for water is challenging as urban water shortages increase (UNESCO, 2020; Gatica & Cytryn, 2013). The reuse of treated wastewater (TWW) is encouraged as a practical solution for water shortage (Becerra-Castro *et al.*, 2015). TWW has extensive applications, namely: irrigation, non-potable urban uses, industrial water use, surface water and groundwater recharge. Of the several demands for freshwater, irrigation accounts for up to 70% of water abstracted from rivers and groundwater (UNESCO, 2020). The predicted increase in the global human population to 9.7 billion in 2050 will lead to an increase in water requirement for agricultural and food production purposes (Zammit *et al.*, 2020; Hong *et al.*, 2018). Thus, TWW is most frequently used for agricultural irrigation in high water-stress areas with strong agricultural sectors such as the southern European countries, southwestern United States, Australia, and Israel (Zammit *et al.*, 2020). Moreover, the reuse of TWW in agriculture is promoted because it has relatively higher nutrients content which can increase crop yields and reduce fertilizer requirements than groundwater, depending on the level of treatment (Hong *et al.*, 2018). Also, TWW reuse helps preserve the freshwater ecosystems by reducing the discharge of nutrient-rich effluent from wastewater treatment plants (WWTPs). Thus, preventing eutrophication and algal blooms (Gatica & Cytryn, 2013).

Conventional wastewater treatments (activated sludge, oxidative ditch, rotatory biological contactors etc.) involve the use of physical, chemical, and biological processes to remove contaminants from wastewater (Rizzo *et al.*, 2013). Biological treatment operations that remove biodegradable materials generate sewage sludge (also called biosolids) which is a major by-product from WWTPs (McLain *et al.*, 2017; Bondarczuk *et al.*, 2016). The exact

composition of different types of sludges depends on the wastewater treatment process. Nevertheless, they all are rich in organic and inorganic plant nutrients (Parsai *et al.*, 2018; Oun *et al.*, 2014). In line with resource recovery, a high amount of sewage sludge produced during wastewater treatment is widely used in agriculture as fertilizer. In the USA, about 60% of biosolids produced annually are applied to land (McLain *et al.*, 2017). Moreover, the use of sewage sludge as a soil amendment may be an important disposal alternative to landfilling and incineration.

However, WWTPs are known as hotspots of antibiotic resistant bacteria (ARB) and antibiotic resistance genes (ARGs) (Amarasiri *et al.*, 2019; Rizzo *et al.*, 2013). WWTPs receive wastewater with a considerable amount of antibiotics, metals, and other selective agents that favour the selection of ARB and propagation of ARGs (Hong *et al.*, 2018; Li *et al.*, 2017). Consequently, TWW and biosolids may contain antibiotics, ARB and ARGs. Many studies have shown the presence of non-regulated chemical and biological contaminants such as pathogens, antibiotics, ARB and ARGs in biosolids and TWW (Alygizakis *et al.*, 2020; Parsai *et al.*, 2018; Al-Jassim *et al.*, 2015; Rizzo *et al.*, 2013; Guo *et al.*, 2012; Munir *et al.*, 2011; Munir & Xagoraki 2011). Table 1 shows that concerning levels of ARGs are typically detected in effluents and biosolids regardless of the degree of treatment process employed. Sharma *et al.* (2016) discussed the role of different wastewater treatment strategies in reducing the spread of AR and gave a comprehensive list of the ARGs levels detectable after treatment. Chemical disinfection processes such as chlorination, ozonation and Fenton oxidation were reported to achieve 1.65–2.2, 0.60, and 2.42–3.38 log- reduction of selected ARGs, respectively (Sharma *et al.*, 2016; Zhuang *et al.*, 2015). However, the concentrations of the disinfectants at which these removals were achieved are several folds greater than doses typically used in WWTPs. For instance, the aforementioned ARGs removal achievable

by chlorination was at a chlorine dose of 40 mg/L (Sharma *et al.*; 2016); whereas, 5 to 20 mg/L is typically used in wastewater treatment (US EPA, 1999). Also, the reported UV fluence was 500 mJ/cm² while 20 mJ/cm² to 100 mJ/cm² is typically used in WWTPs (Michael-Kordatou *et al.*, 2018; Templeton *et al.*, 2009). These information explain why significant concentrations of ARGs are found in biosolids and TWW even after these disinfection strategies (Table 1). Hence, the reuse of TWW and biosolids may facilitate the spread of ARB and ARGs which is a public health concern.

Despite the widespread applications of water reuse in agriculture, a comprehensive regulatory framework that includes AR requirements is still lacking (Hong *et al.*, 2018). Although the reuse of TWW and sewage sludge is considered as solutions to the limited water resource and sludge disposal issues; the potential environmental and human health-related risks are undeniable (Bondarczuk *et al.*, 2016; Oun *et al.*, 2014). It is crucial to analyze the associated environmental and human health risks to assess whether they are safe solutions or not. Several studies have attempted to elucidate the impacts of TWW irrigation and biosolids amendment on antibiotic resistance (AR) development and propagation in the soil microbiome (Hong *et al.*, 2018; Pepper *et al.*, 2018; McLain *et al.*, 2017; Christou *et al.*, 2017; Bondarczuk *et al.*, 2016; Gatica & Cytryn, 2013). In some of these studies, the conclusion based on the research data examined was that biosolids and TWW irrigation had no significant impact on the soil resistome in the long term (Pepper *et al.*, 2018; Rahube *et al.*, 2014; Gatica & Cytryn, 2013). However, it is not enough to assess the AR human health risks and environmental impacts of wastewater irrigation and biosolids amendment based on the soil resistome alone; changes to the food crops resistome and soil properties of such agricultural lands are important (Piña *et al.*, 2020; Amarasiri *et al.*, 2019; Manaia, 2017; Ashbolt *et al.*, 2013). Besides, recent studies have focused mainly on providing guidelines for

AR risks assessment in water and specific environmental matrices (Amarasiri *et al.*, 2019; Larsson *et al.*, 2018; Manaia, 2017; McLain *et al.*, 2017; Bondarczuk *et al.*, 2016; Ashbolt *et al.*, 2013). Also, relatively few studies have provided detailed and holistic framework for AR human health risks assessment during wastewater irrigation and biosolids amendment (Piña *et al.*, 2020; Hong *et al.*, 2018).

This paper discusses issues that relate to antibiotics and AR determinants accumulation in agricultural lands and crops following TWW irrigation and biosolid amendment. Also, it summaries the issues discussed in recent studies on human health- related risk assessment of ARB and ARGs (Amarasiri *et al.*, 2019; Larsson *et al.*, 2018; Manaia, 2017; McLain *et al.*, 2017; Bondarczuk *et al.*, 2016; Berendonk *et al.*, 2015; Ashbolt *et al.*, 2013). Moreover, we outlined specific steps for estimating potential AR human health risks and identified research gaps that are relevant to a comprehensive AR risks assessment during wastewater reuse and biosolids amendment in agriculture. To achieve this, research papers that examined wastewater reuse, land disposal of biosolids, the fate of ARB and ARGs in water and soil matrices, and related areas were reviewed to answer the following questions: a) How do wastewater reuse and biosolids impact soil resistome? b) What are the effects of reclaimed water and biosolids on soil properties, microbial communities and plants? c) How can we assess human-health risk due to AR exposure from TWW irrigated soil and biosolid-amended soil with considerations of antibiotics and other AR co-selecting agents? Consequently, a framework for conducting AR human health-related risk assessment during wastewater reuse and biosolid amendment was suggested. Finally, research gaps in current knowledge that are relevant to a comprehensive AR risks assessment were identified.

131 **Table 1: Representative ARGs concentrations (tetracycline and sulfonamide) commonly found in effluents (or TWW) and**
 132 **biosolids**

Treatment Method	Target ARGs	Biosolids (copies/g)	Effluent (copies/mL)	References
Reverse Osmosis filtration	<i>sul1, sul2,</i> <i>tetB, tetG,</i> <i>tetX,</i>	—	1.18×10^3 - 3.86×10^3	Lu <i>et al.</i> , 2020
Membrane bioreactor	<i>tetM</i> <i>Sul1</i>	—	$\sim 10^4$ $\sim 10^5$	Helmecke <i>et al.</i> , 2020
Activated Sludge & Chlorination	<i>tetO</i> <i>tetQ</i> <i>tetW</i> <i>tetH</i>	9.7×10^4 8.7×10^4 1.8×10^5 5.6×10^4	2.5×10^2 1.6×10^2 4.4×10^2 1.6×10^1	Al-Jassim <i>et al.</i> , 2015

	<i>tetZ</i>	2.2×10^5	5.5×10^3	
Anaerobic, anoxic	<i>tetA, tetB, tetE</i>	$1.5 \pm 2.3 \times 10^9$ to	$1.0 \pm 0.2 \times 10^3 - 9.5 \pm 1.8 \times 10^5$	Mao <i>et al.</i> , 2015
lagoons, activated	, <i>tetG, tetH, tet</i>	$2.2 \pm 2.8 \times 10^{11}$		
sludge and chlorination	<i>S, tetT, tetX, su</i>			
	<i>l1, sul2,</i>			
constructed wetlands,	<i>tetM</i>	2.36×10^8 to 1.50×10^{10}	$2.46 \times 10^4 - 1.21 \times 10^6$	Chen & Zhang 2013
biological aerated and	<i>tetO</i>			
UV disinfection	<i>tetQ</i>			
	<i>tetW</i>			
	<i>sulI</i>	5.39×10^{10} to 5.60×10^{10}	$6.61 \times 10^5 - 3.13 \times 10^6$	
	<i>sulIII</i>			
Activated Sludge	<i>tetO tetW sul1</i>	$1.00 \times 10^8 - 1.78 \times 10^9$	$9.12 \times 10^5 - 1.05 \times 10^6$	Gao <i>et al.</i> , 2012
Anaerobic digestion	<i>tetW</i>	9.53×10^8		Munir & Xagorarakis,
and lime stabilization	<i>tetO</i>	3.15×10^8	—	2011

	<i>sul1</i>	6.04×10^8		
Conventional treatment	<i>tetW</i>	$2.34 \times 10^5 - 2.51 \times 10^7$	BDL* – 4.27×10^3	Munir <i>et al.</i> , 2011
utilities	<i>tetO</i>	$6.31 \times 10^6 - 1.74 \times 10^9$	BDL – 9.12×10^3	
	<i>sul1</i>	$5.62 \times 10^6 - 2.51 \times 10^9$	$2.34 \times 10^4 - 5.62 \times 10^6$	
Activated Sludge	<i>tetC</i>	$3.09 \times 10^8 - 9.33 \times 10^8$	BDL – 1.32×10^4	Zhang <i>et al.</i> , 2009
Chlorination	<i>tetA</i>	$1.23 \times 10^8 - 1.29 \times 10^9$	BDL – 2.14×10^4	
Activated Sludge	<i>tetQ</i>	$2.51 \times 10^8 - 10^9$	$6.31 \times 10^3 - 1.58 \times 10^6$	Auerbach <i>et al.</i> , 2007
& UV	<i>tetG</i>	$3.16 \times 10^8 - 1.58 \times 10^9$	$1.58 \times 10^4 - 7.94 \times 10^5$	

*BDL: Below detection limits. The values presented as a single number are the averages of ARGs in the examined sample (i.e. biosolid or effluent). Values given over a range represent the lowest averages and the highest averages detected for ARGs concentrations examined in samples collected over a study period at the same location or stages of the treatment process.

2. Antibiotic resistance development during wastewater reuse and biosolids application on soils

Conventional wastewater treatment processes are not optimized for the removal of micropollutant (Alygizakis *et al.*, 2020; Helmecke *et al.*, 2020; Rizzo *et al.*, 2013). Moreover, the amplification of AR within the wastewater treatment process is common because biological wastewater treatment is carried out under conditions favourable for microbial growth (McLain *et al.*, 2017; Rizzo *et al.*, 2013). High concentrations of antibiotics, metals, biocides, ARB and ARGs enter the environment through continued use of TWW on soils. Likewise, sewage sludges applied as soil conditioners are treated during one or more treatment processes namely: lime stabilization, thickening, dewatering, drying, anaerobic digestion or composting processes (Munir *et al.*, 2011). The routine application of biosolids improves soil fertility; however, many contaminants like antibiotics, ARB and ARGs may not be removed efficiently. Antibiotics such as tetracycline, sulfamethoxazole, ciprofloxacin, norfloxacin, trimethoprim, and ofloxacin are determined in high concentration in the sludge of different WWTPs (Ekpeghere *et al.*, 2017; Wang *et al.*, 2014). The presence of antibiotics in the biosolids and TWW impact the microbiota and encourages the development of AR in the soil (D'Angelo 2017; Becerra-Castro *et al.*, 2015; Wang *et al.*, 2014).

Naturally, AR in some microorganisms is an evolutionary process that increases their chances of survival against other microbes (Pepper *et al.*, 2018; Calero-Cáceres & Muniesa, 2016). Moreover, undisturbed soils are known as a reservoir of AR in environmental microbes because soil microorganisms are the source of some antimicrobial agents (Pepper *et al.*, 2018; Manaia, 2017; Gatica & Cytryn, 2013). Studies have shown that the ARGs pool in the environment is probably larger than the gene pool observed among pathogens (Bondarczuk *et*

159 *al.*, 2016). In support, the findings of Negreanu *et al.* (2012) showed that ARB and ARGs
160 present in TWW irrigated soils were sometimes lower than in the freshwater irrigated soils or
161 pristine soil. Also, there are suggestions that ARB that enter the soils from the TWW are not
162 able to survive in the soil environment and cannot contribute significantly in the soil
163 microbiota (McLain *et al.*, 2017). Thus, it is easy to think that the release of the relatively
164 smaller ARGs pool in biosolids and TWW should not be a concern. Nevertheless,
165 environmental AR reserve has been observed to contribute to the development of AR in
166 clinical settings (Bondarczuk *et al.*, 2016; Gatica & Cytryn, 2013; Forsberg *et al.*, 2012). The
167 introduction of human pathogens in soil via TWW irrigation and biosolids amendment could
168 promote the acquisition of new ARGs in clinically relevant microorganisms. The
169 metagenomics study conducted by Forsberg *et al.* (2012) showed that horizontal transfer of
170 ARGs between soil and clinical isolates is possible. Also, environmental microorganisms can
171 acquire clinically important genes from pathogens which evolved under a strong selective
172 pressure (Bondarczuk *et al.*, 2016). AR is passed from one microorganism to another by
173 either vertical gene transfer (VGT) and horizontal gene transfer (HGT) (Figure 1).

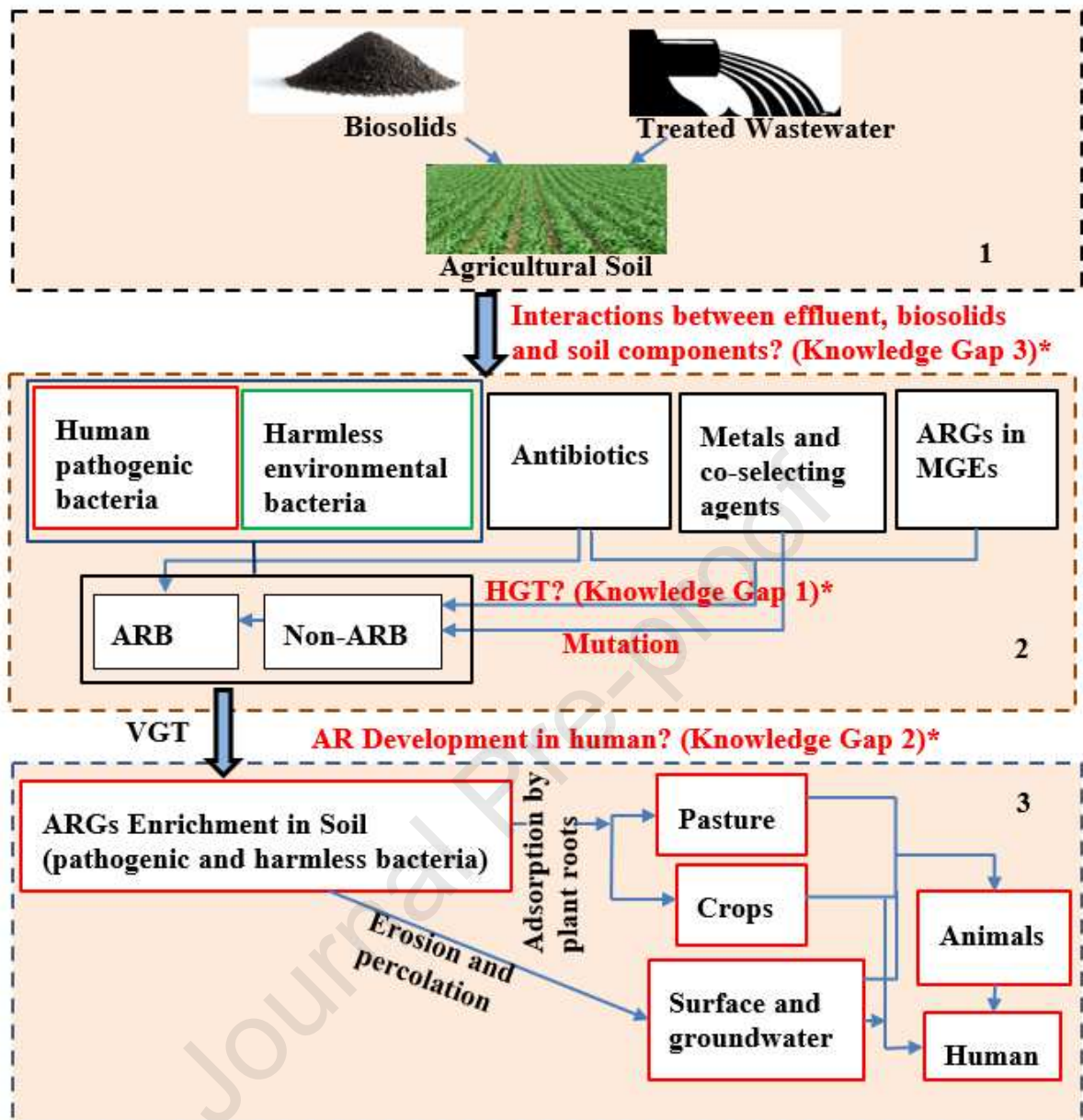


Figure 1: The enrichment of AR in human pathogenic bacteria during TWW irrigation and biosolids amendment.

1. The release of antibiotic resistance (AR) determinants into agricultural soil from treated wastewater and biosolids.
 2. The interactions between antibiotics and co-selection agents on the AR development in soil microorganisms. Selective pressure, mutation, and horizontal gene transfer (HGT) are presented as the mechanisms for AR development in human pathogenic bacteria.
 3. AR dissemination routes to human. Boxes outlined in red show resistome of human health concerns.
- * The question marks represent AR dissemination pathway for which data needed for quantitative AR risk assessment are limited. A detailed description of the identified gaps is given in Table 2.

VGT refers to the transfer of genetic information from parent cells to offspring (Berendonk *et al.*, 2015; Keen & Montforts, 2012). This is rapidly driven by the selective pressure of

187 antibiotics used in therapeutic settings (Michael-Kordatou *et al.*, 2018). The use of TWW and
188 biosolids in agriculture results in the introduction of subtherapeutic amounts of antibiotics in
189 the environment. Subtherapeutic antibiotic concentrations are known to promote the
190 development of AR in microorganisms (Chang *et al.*, 2015). In Sir Alexander Fleming's
191 words – the Scottish Bacteriologist who discovered penicillin, “It is not difficult to make
192 microbes resistant to penicillin in the laboratory by exposing them to concentrations not
193 sufficient to kill them, and the same thing has occasionally happened in the body.” (Nobel
194 Media, 2020). Antibiotics exposure as a selective pressure for ARGs occurrence is a crucial
195 but often underplayed aspect of wastewater reuse in agricultural irrigation. Among the
196 different classes of antibiotics, tetracycline and sulfonamide are widely used to treat human
197 and animal diseases. Their relatively high use in agriculture is because they improve
198 productivity by inducing prophylactic and therapeutic effects (Gao *et al.*, 2012). The
199 antibiotic use in an environment is the major factor that influences the rate of AR
200 development and transmission amongst microorganism in that environment (Chang *et al.*,
201 2015). Studies showing a correlation between antibiotic use and the development of
202 antimicrobial resistance in pathogenic and harmless bacteria have been documented
203 (Berglund *et al.*, 2015; Gao *et al.*, 2012, Pruden *et al.*, 2012). Tetracycline and sulfonamide
204 resistance genes have attracted attention in different studies as they were frequently detected
205 in diverse ranges of concentration in wastewater and sludge (Xu *et al.*, 2015; Laht *et al.*,
206 2014; Miller *et al.*, 2013; Gao *et al.*, 2012; Pruden *et al.*, 2012) (Table 1). Christou *et al.*,
207 (2017) conducted a study to examine the concentrations of five tetracyclines in the
208 rhizosphere soil that was irrigated by reclaimed wastewater for a long time. The total
209 concentration of tetracycline was in the range of 12.7 to 145.2 µg/kg. In another study on
210 biodegradability of pharmaceutical compounds in agricultural soils (carrot crop) irrigated

with TWW, the concentrations of sulfamethoxazole in soil was 0.12 to 0.28 $\mu\text{g/kg}$ (Grossberger *et al.*, 2014). These studies buttress that soil irrigated by reclaimed wastewater accumulates antibiotics in concentrations several folds higher than the antibiotic concentration in the wastewater (Christou *et al.*, 2017; Grossberger *et al.*, 2014). This residual antibiotics leads to the exposure of the soil microbes to different antibiotics (Wang *et al.*, 2014; Christou *et al.*, 2017). Thus, these antibiotics impose selective pressure by killing susceptible bacteria while resistant bacteria proliferate (Berendonk *et al.*, 2015).

In HGT, a bacterium without resistance acquires the resistance genes from mobile genetic elements (MGEs) such as plasmids, integrons, and transposons (called resistant determinants) (Figure 1). HGT mediated processes include conjugation, transduction and natural transformation. Conjugation is a process in which DNA is passed from a cell (the donor) to another cell (the acceptor) through direct cell-to-cell contact; transduction is the introduction of ARGs into microbial cells by bacteriophages; and natural transformation refers to the process in which competent microbes take up free (extracellular) DNA from their surroundings (Keen & Montforts, 2012). The ability of many antibiotic-susceptible bacteria to incorporate ARGs into their cells via HGT makes ARGs persistent in the environment (Lu *et al.*, 2020; Yu *et al.*, 2018; Matyar, 2012). Intact resistance determinants within the bacteria cell debris in TWW and biosolids can confer resistance to bacterial populations downstream via HGT. Also, it has been reported that this mechanism of ARGs transfer is accelerated by environmental stressors such as biocides, toxic metals, and nanomaterial (Zhang *et al.*, 2018; Qiu *et al.* 2012; Soumet *et al.* 2012) (Figure 1). HGT is by far the most important route by which ARGs are transferred from environmental and (or) commensal ARB to antibiotic-susceptible human pathogens (Chang *et al.*, 2015). TWW and biosolids contain highly mobile and versatile mobile genetic elements (MGEs) that contribute to AR propagation far beyond

intrinsic elements that are chromosomally harboured in many native soil bacteria (Gatica & Cytryn, 2013). A study conducted by Zammit *et al.* (2020) revealed that secondary effluent, heterogeneous photocatalysis TWW and chlorinated TWW resulted in an increase in the level of class 1 integrons (*intI1*) in the soil from 1.4×10^4 copies/g before irrigation to 3.3×10^5 copies/g, 1.6×10^5 copies/g and 6.0×10^4 copies/g, respectively, after irrigation over a period of 37 days. Thus, the practice of using TWW in irrigation and biosolids amendment can make ARB and ARGs pseudo-persistent in soils. Furthermore, studies have shown that ARGs may persist longer in the soil than antibiotics. For example, Wang and his research group studied soils of six public parks which were irrigated by the reclaimed wastewater. They observed that there was no antibiotic pressure but sulfonamide resistance genes (*sul1* and *sul2*) persisted in the soil. In like manner, Zhu *et al.* (2018) observed enrichment of ARGs in the soil near a pharmaceutical industry in China whereas the concentrations of antibiotics were below detectable limits. These results indicated that ARGs are more persistent in the environment than antibiotics (Zhu *et al.*, 2018; Wang *et al.*, 2014).

3. Impact of changes in soil properties and microbial diversity on the soil resistome during TWW irrigation

TWW is rich in micro- and macronutrients such as nitrogen, phosphorus, iron, manganese, potassium, calcium, and magnesium that can enhance the fertility of the soil (Gatica & Cytryn, 2013). However, TWW irrigation can result in changes in abiotic and biotic factors of the soil. Changes in the soil abiotic factors result in changes in soil microbiological properties (biotic factor). Abiotic factors that result in changes in microbiological properties of soil include pH, metal pollutant, antibiotic concentration, nutrients, and salinity (Becerra-Castro *et al.*, 2015; Gatica & Cytryn, 2013; Kiziloglu *et al.*, 2007). Environmental stressors

such as changes in pH, salinity, metal concentration etc., are known to induce cross-adaptive responses in organisms – a phenomenon in which sub-lethal exposure of bacterial cells to a biocide results in increased resistance to another lethal agent (Li *et al.*, 2017; Becerra-Castro *et al.*, 2015; Rattan *et al.*, 2005).

Studies have shown that irrigation of soil with TWW results in considerable changes in the soil pH (Becerra-Castro *et al.*, 2015; Rousk *et al.*, 2010; Rattan *et al.*, 2005). pH changes have been observed to increase the minimum inhibitory concentration (MIC) of resistant bacteria and induce antimicrobial resistance in formerly susceptible bacteria. The study of McMahon *et al.* (2007) revealed that the AR response of *E. coli*, *Salmonella* and *Staphylococcus aureus* was significantly altered after these bacteria were exposed to sublethal environmental stresses (e.g. pH). The MICs of the bacteria were found to increase by at least 2-folds while greater than 4-fold increases were detected for some antibiotics after the exposure of the bacteria to reduced pH. Consequently, pH changes in soil can foster the spread and development of AR. According to Rousk *et al.* (2010), pH influences the abundance and composition of bacteria in the soil and there is a positive relationship between the quantity and diversity of bacteria at soil pH in the interval pH 4–7. Also, pH impacts the concentration of metals in soil by affecting their solubility. For instance, Rattan *et al.* (2005) observed an increase in the concentration of free metals in soil irrigated with TWW. This increase in metal concentration resulted from a decrease in soil pH (Kiziloglu *et al.*, 2007; Rattan *et al.*, 2005).

WWTPs receive considerable amounts of metals and TWW irrigation results in the release of metals into the soil (Becerra-Castro *et al.*, 2015; Seiler & Berendonk, 2012 ;). Metals in the soil can result in the co-selection of metal resistance bacteria (MRB) and ARB. A study by Li

et al. (2017) showed a relationship between metals resistance genes (MRGs) and AR. In the study, they observed that metal pressure promotes AR by co-selecting for ARB and MRB. Several other studies on the co-selection of MRB and ARB exist (Li et al. 2017, Pal et al., 2017, Pal et al., 2015; Seiler & Berendonk, 2012; Wardwell et al., 2009; Akinbowale et al., 2007). Co-selection mechanisms include co-resistance and cross-resistance (Seiler & Berendonk, 2012). In co-resistance, different resistance determinants are present on the same genetic element (Zhang et al., 2018; Seiler & Berendonk, 2012). For example, Hasman and Aarestrup, (2002) discovered *Enterococcus faecium* isolated from farm animals harboured *tcrB* gene, macrolide (*ermB*) and glycopeptide resistance (*vanA*) genes with a genetic linkage of Cu resistance. Some ARGs exhibit resistance to both antibiotics and metals, known as cross-resistance genes. For instance, *tetL*, *merE*, and *oprD* genes are resistant to both antibiotics and metals (Zhang et al., 2018). Efflux pump is an AR mechanism that is a major driver of cross-resistance (Seiler & Berendonk, 2012; Baker-Austin et al., 2006). Baker-Austin et al. (2006) gave a comprehensive review on cross-resistance mechanisms. Metals are environmental stressors that promote the dissemination of AR by accelerating HGT (Zhang et al., 2018; Li, et al., 2018; Becerra-Castro et al., 2015; Qiu et al. 2012; Soumet et al. 2012). Zhang et al. (2018) discovered that Cu(II), Ag(I), Cr(VI), and Zn(II) at environmentally-relevant and sub-inhibitory concentrations promoted conjugative transfer of ARGs between *E. coli* strains. High metal concentrations in soil environments not only facilitate co-selection processes but also increase the level of tolerance of microorganisms to antibiotics due to co-regulation of resistance genes (Seiler & Berendonk, 2012).

The concentrations of dissolved inorganic salts in TWW is relatively higher than those in other freshwater sources (Gatica & Cytryn, 2013; Kiziloglu et al., 2007). Salinity is well documented to limit plant growth by imposing hyperosmotic and ion toxicity to the soil

(Becerra-Castro *et al.*, 2015, Kiziloglu *et al.*, 2007). Moreover, soil salinity impacts the community diversity and activity of the soil (Becerra-Castro *et al.*, 2015; Gatica & Cytryn, 2013; Kiziloglu *et al.*, 2007). High salt content imposes oxidative stress to soil microorganisms (Becerra-Castro *et al.*, 2015). Oxidative stress is known to contribute to conjugative gene transfer by damaging bacterial cell membranes and enhancing the expression of mating pair formation genes as well as depressing the expression of regulatory genes that control the conjugative transfer (Qiu *et al.* 2012).

The presence of residual nutrients in TWW is one of the advantages of its use in irrigation. Nevertheless, higher nutrients contents in WWTPs are more beneficial for HGTs of ARGs since substrate (carbon, nitrogen and phosphate) promote bacteria growth (Yuan *et al.*, 2018; McLain *et al.*, 2017). Commonly, nutrients concerns in TWW irrigation involve the disturbance of soil microbial communities (Becerra-Castro *et al.*, 2015). Now, recent studies have provided positive correlation between ARGs and nutrients concentrations. Li *et al.* (2018) reported that ammonium had a positive correlation with ARGs and significant effects on the ARGs distribution in the sediments. Similarly, a study by Zhang *et al.* (2020), showed that an increase in nutrients levels (phosphate, total phosphorus, ammonium, nitrate, nitrite, total nitrogen, silicate and total organic carbon) in water generally correlated positively with the abundances of ARGs; although, a negative nutrient correlation was found with the abundances of 16S rRNA genes. Since nutrients and ARGs exist nearby in TWW and TWW irrigated soil, the impact of nutrients in TWW on the soil resistome may be undeniable.

Most concerns regarding wastewater reuse for irrigation are related to the presence of pathogenic microbes in the TWW. However, the microbiological components of the soil that might be altered by the increase of soil microbial biomass due to TWW irrigation are rarely

considered. Several studies have proven that soil irrigation with TWW has improved microbial activity (Ibekwe *et al.*, 2018; Becerra-Castro *et al.*, 2015). Nevertheless, changes in the microbial diversity of the soil due to this practice have been confirmed (Ibekwe *et al.*, 2018; Rutai *et al.*, 2015; del Mar Alguacil *et al.*, 2012). For instance, the effect of treated urban wastewater irrigation on fungi diversity and soil microbial activities was studied by del Mar Alguacil *et al.* (2012). Based on this study, fungi diversity was higher in soil irrigated by freshwater but microbial activities of soil irrigated by TWW were much more than the soil irrigated by freshwater. Ibekwe *et al.* (2018) recorded that in addition to *Bacillus* and *Mycobacterium* in soils irrigated with synthetic freshwater (SFW), *Clostridium* was detected in TWW irrigated soils. Also, the study found higher levels of *Acinetobacter*, *Legionella*, and *Pseudomonas* in TWW irrigated soils than in SFW irrigated soils. *Acinetobacter* and *Pseudomonas* are amongst the list of the 12 ARB families which are considered to pose great threats to human health (WHO, 2017). This change in microbial diversity results due to the selection for microorganisms that can better survive new environmental conditions created by the irrigation water (del Mar Alguacil *et al.*, 2012). As ARB are known to survive harsh environmental conditions with high competition, it suffices to say that their population will be considerably favoured in TWW irrigated soils (D'Angelo 2017; Li *et al.*, 2017; Becerra-Castro *et al.*, 2015). Also, Weidner *et al.* (2015) elucidated that changes in the microbial diversity of the soil affect microbial community functioning and interaction. This microbial interaction plays a key role in major activities, such as gene expression patterns in microorganisms and soil-borne pathogens suppression (Rutai *et al.*, 2015; Weidner *et al.*, 2015).

4. Effects of TWW irrigation and biosolid reuse on plants

Biosolids and TWW contain elevated concentration of antibiotics, ARB and ARGs (McLain *et al.*, 2017; Munir *et al.*, 2011). Studies on the fate of antibiotics are motivated by two main concerns; antibiotics in the environment may contribute to the abundance of antibiotic resistant pathogens, and the ecological consequences of antibiotic contamination that may enter the human food chain (Ben *et al.*, 2019; Clarke & Smith 2011). Studies have shown that antibiotics can be uptaken by plants in soils irrigated by wastewater or amended by biosolids (Christou *et al.*, 2017; Azanu *et al.*, 2016; Franklin *et al.*, 2016; Pan *et al.*, 2014; Malchi *et al.*, 2014; Clarke & Smith, 2011). Some studies have reported the translocation of some antibiotics (e.g. sulfamethazine) into the aerial plant components from the soil matrix (Christou *et al.*, 2017; Malchi *et al.*, 2014; Clarke & Smith, 2011)). For instance, Franklin *et al.* (2016) found that sulfamethoxazole was concentrated within the grain ($0.64 \pm 0.37 \mu\text{g/kg}$) of wheat plants grown on fields irrigated with TWW. Also, Malchi *et al.* (2014) detected sulfamethoxazole in sweet potatoes and carrot roots (edible parts) ($0.05\text{--}0.24 \mu\text{g/kg}$ wet weight) grown on soil irrigated with TWW. These show that the accumulation of antibiotics in the edible parts of crop plants is feasible. Antibiotic uptake depends on the crop species, soil type, physicochemical properties of antibiotics and concentration in TWW and biosolids (Cerqueira *et al.*, 2019; Christou *et al.*, 2017). Tetracycline and sulfonamide are the two most common classes of antibiotics in the plants which have been detected. Tetracycline and sulfonamide antibacterial have been identified and quantified in radish, rape, celery, cilantro, white cabbage, water spinach, corn and rice (Christou *et al.*, 2017; Pan & Chu, 2017; Azanu *et al.*, 2016; Franklin *et al.*, 2016; Pan *et al.*, 2014; Malchi *et al.*, 2014; Clarke & Smith, 2011). Although these antibiotics in plants are in minute concentrations that may not pose immediate health risks to human, other unregulated pollutants in TWW and biosolids may

accumulate simultaneously in food crops, increasing risks (Pan & Chu, 2017; Pan *et al.*, 2014).

Recent studies have confirmed the impacts of biosolid amendments on the abundance of ARGs in food crops grown on fertilized fields (Zhang *et al.*, 2019, Cerqueira *et al.*, 2019; Yang *et al.*, 2018; Rahube *et al.*, 2014). Rahube *et al.* (2014) reported that vegetables (carrots, radishes, lettuce, cucumber, tomato, and pepper) that were harvested from biosolid fertilized plots in the biosolid application season had a higher abundance of several MGEs and ARGs than vegetables harvested from the control plots. It is noteworthy that ARGs encoded in MGEs are one of the drivers of AR dissemination between antibiotic resistant and susceptible microorganisms (Amarasiri *et al.*, 2019; Gatica & Cytryn, 2013). The same study revealed that vegetables harvested from biosolid fertilized plots the following year after biosolid fertilization showed no difference in the ARGs abundance in vegetables from the control plots (Rahube *et al.*, 2014). This implies that the contribution of TWW irrigation and biosolids fertilization on the abundance of ARB and ARGs in food crops or plant depends on the waiting period between the biosolids application and harvest (McLain *et al.*, 2017; Rahube *et al.*, 2014). Also, Cerqueira *et al.* (2019) analyzed the distribution of *sul1*, *tetM*, *qnrS1*, *blaCTX-M-32*, *blaOXA-58*, *mecA*, *blaTEM*, and *intI1* in soil, rhizospheric soil, roots and leaves of tomato, lettuce and broad beans crops grown on nine commercial farmland across different geographical locations. They observed that the crop type was the main determinant of the ARGs distribution in the crop. The authors proposed that soil amendments were the main driver of ARGs load on the edible parts of the crops and not the irrigation water (Cerqueira *et al.*, 2019). These studies buttress that the consumption of crops fertilized with biosolids and irrigated by TWW is a potential route of exposure to antibiotics, ARGs and ARB. Consequently, significant issues like chemical and biological contaminants in food

crops and their effect on human health need to be considered while using biosolids and TWW in agriculture.

5. Interactions between common constituents of biosolids and TWW in soil

Antibiotics, ARB, ARGs, metal, organic carbon and pathogens are released to soil during TWW irrigation and biosolids amendment (Piña *et al.*, 2020; Amarasiri *et al.*, 2019; Parsai *et al.*, 2018; Mc Lain *et al.*, 2017; Berendonk *et al.*, 2015; Oun *et al.*, 2014). The assessment of the influence of other contaminants to AR development in the soil environment after biosolids application and(or) TWW irrigation is challenging since there are various contaminants that can intensify or moderate each other's effects (D'Angelo, 2017; Becerra-Castro *et al.*, 2015; Wang & Wang, 2015). A pertinent issue is the co-existence of antibiotics and metal contaminants (D'Angelo, 2017; Hu *et al.*, 2017; Becerra-Castro *et al.*, 2015; Wang & Wang, 2015; Zhao *et al.*, 2013; Wan *et al.*, 2010; Jia *et al.*, 2008; Kong *et al.* 2006). The levels of monovalent metallic ions such as (Na^+ , K^+) and divalent multivalence metal ions (Ca^{2+} , Mg^{2+} , Cu^{2+} , Al^{3+} , and Fe^{3+}) impact the adsorption behaviours of antibiotics in the soil (Wang & Wang, 2015). A well-written review on the impacts of other environmental factors on antibiotic adsorption is given by Wang & Wang (2015).

Studies have shown that when soils are contaminated by both antibiotics and metals simultaneously, metals can increase the adsorption of antibiotics to the soil (D'Angelo, 2017; Becerra-Castro *et al.*, 2015; Zhao *et al.*, 2013; Wan *et al.*, 2010; Jia *et al.*, 2008; Kong *et al.* 2006). This promotion of antibiotics adsorption by metal leads to the enrichment of antibiotics in soils without loss of antimicrobial activity (Becerra-Castro *et al.*, 2015, Chander *et al.*, 2005). The accumulation of antibiotics in the soil contributes to AR selection (D'Angelo, 2017; Becerra-Castro *et al.*, 2015; Wang *et al.*, 2014). Metals such as Cu, Pb and

Cd increase the adsorption of antibiotics in soil (Becerra-Castro *et al.*, 2015; Zhao *et al.*, 2013; Wan *et al.*, 2010; Jia *et al.*, 2008). A recent study on the potential risks of the presence of antibiotics in biosolid amendments examined the adsorption and desorption of tetracycline using four types of soil amendments (biosolids, poultry manure, wood chip litter, and rice hull litter) at different temperatures (D'Angelo, 2017). The sorption equilibrium constants in municipal biosolids were 20 times higher than other amendments since the concentrations of bound Al^{3+} and Fe^{3+} were higher in municipal biosolids. Also, the study showed that the sorption of tetracycline significantly increased after treatment with alum. Thus, suggesting that treatment of amendments effectively reduced antibiotic diffusion rates (D'Angelo, 2017). Furthermore, the presence of metals and antibiotics in soil have been observed to have a synergistic effect on reducing the diversity of bacteria. Kong *et al.* (2006) assessed the effects of a widely used veterinary antibiotic (oxytetracycline), Copper (Cu) and a combination of both. Based on this study, oxytetracycline and Cu significantly diminished the functional diversity of soil microbial community, and the combination of them resulted in an extra reduction (Kong *et al.*, 2006). Thus, the interactions between contaminants are crucial in assessing the role of TWW irrigation and biosolids amendment in AR dissemination.

6. Human health risk assessment during wastewater reuse

The reported occurrences of ARB, ARGs (Table 1) and their co-presence in environmental matrices (water and soil) suggest the possibility of risks to humans (Amarasiri *et al.*, 2019; Tong *et al.*, 2019; Zhang *et al.*, 2018; Sharma *et al.*, 2016; Al-Jassim *et al.*, 2015; Mao *et al.*, 2015; Munir & Xagorarakis, 2011). Numerous researchers across the globe have attempted to understand potential risks due to exposures of antibiotics, ARB and ARGs using quantitative approaches such as: quantitative microbial risk assessment (QMRA), molecular techniques

and culture-based experiments, and epidemiological studies (Amarasiri *et al.*, 2019; Hong *et al.*, 2018; McLain *et al.*, 2018; Manaia, 2017; Bondarczuk *et al.*, 2016; Ashbolt *et al.*, 2013). However, due to absenteeism of dose-response models for ARB and ARGs, calculating the risk of AR infection is difficult. This section summarizes the steps involved in human health risk assessment during TWW irrigation and biosolid amendment (Figure 2). It also includes some knowledge gaps in conducting a comprehensive human health risk assessment due to the usage of TWW and bio-solids in agricultural practices (Figure 2).

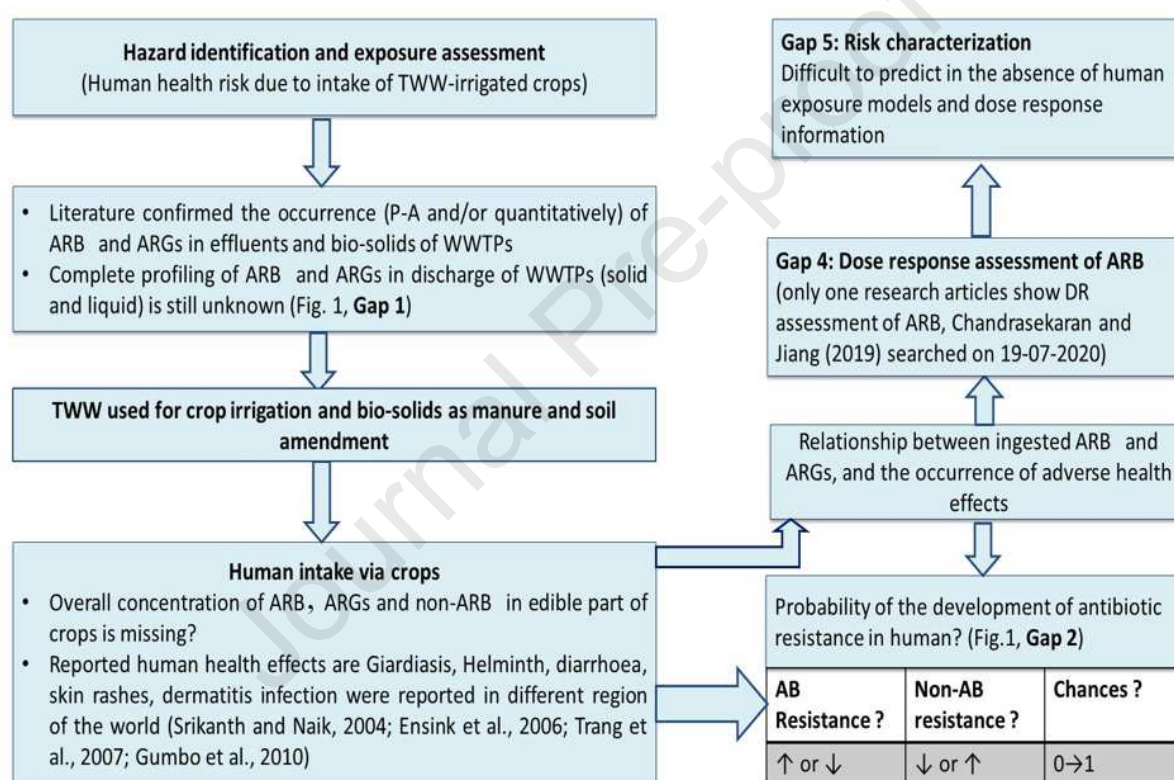


Figure 2: Human health risk assessment due to the consumption of TWW-irrigated crops. Figure summary: TWW and biosolids contain a mixture of ARB, ARGs and non-ARB. The flow diagram shows risk assessment steps and associated knowledge gaps. DR = dose response, AB = antibiotics, TWW = treated wastewater, ARB = antibiotic resistant bacteria, ARGs = antibiotic resistance genes.

6.1 Hazard identification

Studies have reported that the co-presence of antibiotics, pathogenic bacteria, environmental bacteria and ARGs in the environment might lead to the development and dissemination of AR in the bacterial community (Ben *et al.*, 2019; Berglund, 2015; Rahube *et al.*, 2014; Ashbolt *et al.*, 2013). As discussed in Section 4, the translocation of ARB and ARGs to the edible parts of food crops is possible (Zhang *et al.*, 2019, Cerqueira *et al.*, 2019; Yang *et al.*, 2018; Rahube *et al.*, 2014). Consequently, pathogenic and human commensal ARB may enter into, inhabit and multiply in the human body. ARB and ARGs are microbiological hazards related to AR human health risks assessment during TWW irrigation and biosolids amendment (Ashbolt *et al.*, 2013; Snary *et al.*, 2004). The definition of ARB and ARGs as hazards reflects the potential for resistant bacteria to cause adverse health effects and the potential for HGT between microorganisms resulting in AR development. Pathogenic ARB can increase the risks of infection. Moreover, commensal ARB may harbor crucial virulence genes and cause diseases, or they could transfer virulence genes to another commensal human microbiota (Ben *et al.*, 2019; Manaia, 2017; Li *et al.*, 2015). Also, ARGs can initiate AR development in non-AR pathogenic bacteria; although, the chance of ARGs transfer leading to an AR infection relies upon the presence of recipient pathogenic bacteria (Ben *et al.*, 2019; Manaia, 2017; Ashbolt *et al.*, 2013).

Antibiotics are infamously recalcitrant to conventional biological treatment in WWTPs. Thus, they are constantly released into agricultural fields receiving TWW (Pina *et al.*, 2020). It is well-known that sulfonamides and macrolides (azithromycin, clarithromycin, and erythromycin) classes of antibiotic undergo incomplete removal in WWTPs with the additional formation of metabolites by conjugating with human excreta during biological treatment processes in WWTPs. Such antibiotics deserve special attention when considering the possible hazard emissions via TWW (Pina *et al.*, 2020, Ben *et al.*, 2019). Due to this,

macrolides were included in the latest version of the EU Watch List of Chemicals from the Water Framework Directives (EU 2015/495). Other antibiotics such as fluoroquinolones and tetracyclines are mostly released to soils through fertilization with bio-solids due to significant accumulation in sewage sludge. Because of their continuous release into the environment, these antibiotics are considered as pseudo-persistent contaminants of emerging concern (Kim *et al.*, 2005; Giger *et al.*, 2003). Moreover, several classes of antibiotics (chloramphenicol, sulfonamides, fluoroquinolones, and lincosamides) are reported to have high bioconcentration factors (BCFs) and tend to accumulate in crops and plants grown with biosolid amended soil and TWW (Pina *et al.*, 2020; Christou *et al.*, 2017; Christou *et al.*, 2016; Franklin *et al.*, 2016; Paltiel *et al.*, 2016; Riemenschneider *et al.*, 2016; Pan *et al.*, 2014).

6.2 Exposure assessment

The routes of exposure to these identified hazards include (i) inhalation of ARB/ARGs/antibiotics in aerosols produced on agricultural lands during irrigation and soil amendment, (ii) absorption through the skin following contact with TWW, biosolids or contaminated crop and (iii) ingestion (mostly via consumption of contaminated crops) (Ben *et al.*, 2019; Amarasiri *et al.*, 2019; Manaia, 2017). Thus, information about the presence of antibiotics, ARB, and ARGs in different environmental systems is required so that all hazards can be systemically incorporated during the exposure assessment step. Several efforts have been made to quantify the antibiotic exposure risks of humans following the consumption of crops irrigated with TWW and(or) grown on biosolid amended soil. For instance, Marsoni *et al.* (2014) studied the potential adverse effects of lincomycin and ofloxacin along the food chain through lettuce consumption. The reported annual consumption of lincomycin and

ofloxacin were 26 µg and 16 µg, respectively which correspond to 0.005 and 0.008% of the medical dose equivalent.

To explore more on antibiotics hazard exposure risks, researchers have assessed the annual or daily acquaintance of humans to antibiotics and reported in the form of medical hazard quotient (HQ) and(or) threshold of toxicological concern (Pina *et al.*, 2020; Prosser and Sibley, 2015; Malchi *et al.*, 2014; Pan *et al.*, 2014;). By using similar approaches, Christou *et al.*, 2017 analysed the effect of sulfamethoxazole (24.8-55.3 ng/L), trimethoprim (22.2-73.2 ng/L) and diclofenac (25.8-49.6 ng/L) on adults and infants due to the consumption of tomato picked from plants irrigated with tertiary TWW for three consecutive years. They noted that the hazard quotient value was significantly low <0.015 (note that only HQ>1 indicates risk and a cause of concern). The reported hazard quotient indicates that usage of wastewater in irrigation caused no human risk. However, the calculated tomato BCFs were high for diclofenac in the second (108) and the third year (132) of the experimental period. Similarly, Prosser and Sibley (2015) evaluated the health risks due to the consumption of plants accumulated with antibiotics and other pharmaceuticals and reported the HQ to be < 0.1, indicating no risk. Although the aforementioned studies considered that the majority of human received a minimum or no health effect antibiotic dose after the consumption of crops irrigated with TWW, none of them incorporated exposures of ARGs. Moreover, opposite views have been presented in other studies with regards to other pharmaceuticals exposures. For instance, a study conducted by Paltiel and his coworkers (2016) proved the presence of carbamazepine and its metabolites in the urine samples of personnel who consumed reclaimed wastewater treated vegetables. This study indicated the inadvertent intake of antibiotic, other pharmaceuticals, and their by-products at high or low concentration. The presence of antibiotics in urine and blood may provoke physiological and possibly opposed

responses depending upon the exposure concentration as well as the continuous exposure to a cocktail of antibiotics. Despite the availability of the enormous number of studies, the mechanistic understanding of antibiotics, ARB and ARGs uptake by crop plants remains uncertain including their human health risk framework (Miller *et al.*, 2016). The foregoing indicates that there is a need for incorporating both antibiotics, ARB and ARGs from different environmental matrices in the exposure assessment stage.

6.3 Dose-response (DR) assessment

Due to the unavailability of dose-response data for ARB and ARGs, none of the researchers cited in Section 6.1 performed the QMRA analysis. However, they tried different approaches to address the issue. For example: in case of antibiotics, they used minimum inhibiting concentrations of antibiotics (predicted-no-effect-concentration (PNEC)) to human intestinal flora following the EMEA methodology (EMEA, 1998) as acceptable daily intake value for risk estimation due to exposures of different classes of antibiotics, such as tetracycline, fluoroquinolones and sulfa drugs (Kumar and Xagorarakis, 2010; Kumar *et al.*, 2010). In the case of ARB, Leonard *et al.* (2015) assessed human recreational exposure to ARB (*E. coli*) in coastal bathing waters of Wales and England where they first exposed coastal waters to cefotaxime and ceftazidime antibiotics to determine the prevalence of resistance amongst culturable *E. coli* (Third generation cephalosporins-resistant *E. coli*, 3GCREC) and then estimated the number of 3GCREC ingested during different water-related activities. They indicated the need for conducting experiments for (1) demonstrating the relation of ARB exposure to the colonization of ARB in the human gut, (2) understanding survival of ARB in the human alimentary canal and different factors that affect ARB colonization in the human gut, (3) developing dose-response relationships for colonization of ARB in the human gut. To

understand the implications ARGs exposure from different environmental matrices on human health, it is important to understand how ARGs interact with bacteria in the environment and human. The interaction of ARGs with bacteria can increase the chances of resistance development in bacteria. Once developed, pathogenic type ARB can spread between humans. Berglund (2015) indicated the need for knowing information on concentrations of antibiotics which may initiate resistance and dissemination in bacteria under different environmental conditions. For estimating the risk of exposure to genetic materials, such as ARGs, very few information is available. Non-ARB within the body might acquire resistance with the help of these materials and thus quantification of health risks due to the inadvertent exposure to these genetic materials become important. More efforts are required to elucidate the mechanism of how pathogenic bacteria within human might develop resistance to antibiotics after exposures to ARGs which might affect the recovery after initiation of infection.

Quantitative human health risks due to TWW irrigation and biosolids amendment has not been fully explored yet due to (i) the absence of complete inventory of the presence of antibiotics, ARB and ARGs alone and their multi-mixture and by-products in the effluent of WWTPs (globally, locally and regionally) (Gap 2, Figures 1 and 2) and (ii) the unavailability of exposure assessment and dose-response data (epidemiological studies) related to ARB and ARGs on humans (Ashbolt *et al.*, 2013; Pepper *et al.*, 2018) (Gap 4, Figure 2). More research is needed in this direction, so that risk estimate can represent the actual health risks effects.

While formulating the risk assessment for antibiotics, ARB and ARGs, current parameters need to be adjusted by incorporating the new knowledge of antibiotics toxicity and hazards. Special emphasis should be given to antibiotics (ciprofloxacin and sulfa drugs) which shows genotoxicity and carcinogenicity and those antibiotics whose toxicological thresholds are

extremely low ($2.5 \text{ ng kg body weight}^{-1}\text{d}^{-1}$). Moreover, the existing risk assessment framework undervalues the threats associated with antibiotics transformation products (TPs). There is a chance that antibiotics TPs may accumulate in plants as well as in environmental matrices at a greater extent than the parent products and caused higher toxic effects.

6.4 Dose-response related considerations for quantifying AR-based risks

Till date, only Ashbolt et al. (2013) published a QMRA framework which outlined the complex nature of this type of risk assessment and assumed that the behaviour of ARB is similar to their non-ARB complements in terms of their pathogenicity and follows similar dose-response relationship. However, the response to antibiotic treatment is likely to be different. The greatest uncertainty in Ashbolt's QMRA approach is associated with the assessment of dose-response resulting from human exposure (Haas *et al.*, 1999). For proper risks assessment due to biosolid and TWW on agricultural land, a more complete understanding of pathogen richness, assortment and contagion is essential. No infectious dose-response assessment has been conducted for an ARB pathogen because experimental data have not been collected. Without precise dose-response data for ARB, QMRA cannot be conducted. Similarly, dose-response relationships that explicitly include ARGs in risks estimation are not available in literature. Hence, this needs to be studied in forthcoming research.

Nevertheless, if we assumed that an AR *E. coli* (for example) present in a given volume of TWW follows a non-AR *E. coli* dose-response model, then the probability of infection is calculated using Equation 1 (Beta-Poisson model) (Haas, 2015):

$$\text{Probability of illness } P(d) = 1 - \left(1 + \frac{N}{\beta}\right)^{-\alpha} \quad (1)$$

598 Where, N = Number of organisms ingested per exposure, $\beta = 1.61 \times 10^6$, and $\alpha = 0.1705$

599 Suppose we were to conduct a quantitative AR human health risk assessment due to the
600 consumption of crops irrigated with TWW presented in Table 1, we would face difficulty due
601 to the lack of the following data: (i) the concentrations of antibiotics, ARB and ARGs
602 transferred to the edible crops in this case or other environmental media (Gap 3, Figure 1);
603 (ii) exposure dose for an individual (diet/body weight) and (iii) dose-response related
604 parameters for estimating health risks due to exposures of ARB and ARGs. Moreover,
605 accurate calculation of the risks of infection and illness depends upon the gastrointestinal
606 tract status of humans, self-immunity and host age (Haas, 2015). Thus, holistic studies that
607 examine the accumulation/dissipation of AR determinants in environmental media and
608 provide dose-response information are needed (Gap 2, Figure 1).

609 **7. Identified knowledge gaps and future directions**

610 **7.1 Knowledge gaps and future directions towards quantitative AR risks assessment**

611 The following knowledge gaps were identified towards structuring framework to assess and
612 estimate the risk of human exposure to antibiotics, ARB and ARGs from TWW irrigation and
613 biosolids in agriculture (Table 2).

Table 2: List of identified knowledge gaps for quantitative AR human health risk assessment due to exposures of antibiotics, ARB and ARGs from biosolid and TWW

Gaps	Description	Recommendations
Knowledge Gap 1 (Figure 1)	Very few studies on factors influencing HGT between human pathogenic bacteria and environmental bacteria.	A comprehensive estimation of the human health risk to ARB and ARGs exposure from biosolid and effluent irrigation require more studies that examine the levels of subtherapeutic antibiotics including their by-products, biocides and other co-selection agents in environmental matrices (e.g. soil) that facilitate HGT between environmental ARB and human pathogenic bacteria; and the development of AR in clinically relevant bacterial strains in the water environment (Larsson <i>et al.</i> , 2018; Manaia 2017; McLain <i>et al.</i> , 2017; Bondarczuk <i>et al.</i> , 2016; Ashbolt <i>et al.</i> , 2013). Also, studies that document the rate of ARGs transfer between the environmental bacterium and human pathogenic bacteria are needed. This is useful for an exposure assessment. Furthermore, such information is necessary for setting regulations to limit antibiotics and other co-selection agents in the

		environment.
Knowledge Gap 2 (Figures 1 and 2)	Holistic studies that examine the accumulation/dissipation of antibiotics, ARB and ARGs from irrigation/biosolid application stage to crop consumption stage (humans) in the long term.	Studies that examine the possible exposure of humans to AR determinants through crop ingestion are required. Information about the effects of exposure to antibiotics and AR through ingestion of crops grown on wastewater irrigated and biosolid-amended soil is important to risk estimation (Manaia 2017). Moreover, most of the studies regarding the plant uptake of emerging contaminants (Antibiotics, ARB, and ARGs) were conducted in laboratory scales or at greenhouses. The accumulation of these contaminants in crops at real field condition irrigated by TWW is inadequate. Studies on the topic of sorption or degradation of antibiotics in the soil are limited and most of them used elevated concentrations rather than real environmental concentrations because of limitations in the analysis of minute concentration (Christou <i>et al.</i> , 2017; Becerra-Castro <i>et al.</i> , 2015).
Knowledge Gap 3 (Figure 3)	Limited studies on the influence of soil environmental	Studies that assess the fate of ARGs and ARB in biosolid amended soils with focus on environmental factors (e.g. temperature, salinity) and soil conditions/characteristics (moisture content, organic carbon composition etc.) that promote the dissipation or

1)	conditions on the soil and edible plants resistome.	persistence of ARGs in the environment are needed. Also, extensive studies on conditions that promote reactivation of ARB should be carried out to prevent such occurrences downstream of TWW application. Such studies will promote an understanding of the degradation of ARGs in the environment and factors that influence ARGs degradation. Also, this will provide a useful guide for decision making regarding the application of biosolids on farmland.
Knowledge Gap 4 (Figure 2)	Unavailability of dose-response models for quantifying the risk of infection due to ARB	Enumerating the risk of infection requires information on the likelihood of a person being infected by a given concentration of pathogens (ARB) and this will be done only by the help of ARB specific dose-response models. Nevertheless, risk identification for ARB has been stalled by the lack of suitable dose-response models for ARB. A new systematic experimental approach is required which can capture toxicity studies using mixtures of antibiotics, ARB and/or non-ARB, and ARGs and/or non-ARGs, to obtain dose-response information, and their potential risks. While we await relevant studies in this regard, Equation 1 is suggested for estimating the probability of illness due to ARB exposure.

7.2 Where we are in terms of disinfection standards for reclaimed water/treated wastewater containing ARB and ARGs

Current wastewater treatment practices significantly reduce bacterial and viral concentrations in wastewater (Sharma *et al.*, 2016). However, efficacy depends upon the chosen treatment. Several standards given for reclaimed water by environmental protection and regulatory agencies are class and treatment-based standards (Text T1 in Supplementary Information). These include the United States Environmental Protection Agency Standards for Water Reuse (US EPA, 2012), the Environment Protection Agency Victoria (464.2, Guidelines for Environmental Management: Use of Reclaimed Water) and the World Health Organization Standard for Lawn Irrigation (WHO, 1989 and 2006). The above-stated standards included total and faecal coliforms, helminth eggs and enteric viruses in their monitoring. However, no standard is available for ARB and ARGs monitoring. Recent metagenomic studies of TWW and biosolids have shown that bacterial groups that predominantly host ARGs of clinical relevance and persist within wastewater treatment trains are not always coliforms (Wang *et al.*, 2020; Guo *et al.*, 2017; Li *et al.*, 2015). Bacterial groups such as *Acinetobacter*, *Clostridium*, *Bacteroides*, *Streptococcus* and *Mycobacterium* are frequently identified, all of which are human pathogens (Wang *et al.*, 2020; Guo *et al.*, 2017; Li *et al.*, 2015). Thus, it is our understanding that current indicator organisms used for conventional water quality monitoring may not be adequate for determining the AR status of many environmental matrices. The choice of microorganism groups suitable as ‘indicators’ of AR status needs further considerations. Furthermore, disinfection norms for biosolids are almost non-existent across the world. Overall, future research directions should include ARB and ARGs in their monitoring plans. Field-based research that focus on region-specific ARB and ARGs are

needed. Also, guidelines should be revised based on long-term research work and as antibiotics utilization pattern changes.

8 Closing remarks

The use of TWW for irrigation enjoys benefits. However, the accumulation of antibiotics, ARB and ARGs in the soil and its effect on AR occurrence, soil properties, human and environmental health requires considerations. There is significant uncertainty regarding the fate of antibiotics and ARGs in TWW and biosolids amended soil. It is not enough to assess the impact of TWW irrigation and biosolids amendment on the soil resistome, changes to the resistome of the food crops grown on such agricultural land are important. Hazards linked to the occurrence of antibiotics, ARGs and ARB in food crops and their linkage with TWW irrigation practices are mostly overlooked. Current knowledge cannot deny the fact that ARB and ARGs in the TWW can be transmitted to humans even at low concentration and may cause asymptomatic indelible settlement that is noticed only when wellbeing is negotiated. Therefore, holistic studies that examine the accumulation/dissipation of antibiotics, ARB and ARGs from irrigation/biosolid application stage to crop consumption stage will help a comprehensive human health risk assessment. Also, the interactions between contaminants are crucial in assessing the AR human health-related risk during TWW irrigation and biosolids amendment. HGT is an important route by which ARG is transferred from environmental and(or) commensal ARB to antibiotic-susceptible human pathogens. Hence, more studies on soil conditions that increase the frequency of HGT between the native soil resistome and pathogenic microbes in biosolids and TWW are needed.

Acknowledgements

The authors would like to thank the University of North Carolina Charlotte (USA) and Indian Institute of Technology Delhi (India) for providing financial support for this study.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: